



Department of Electronics and Communication Engineering
Khulna University of Engineering & Technology , Khulna

OPEN ENDED LAB

Project Proposal: Reaction Time Race Game (RTR-Game)

Course Code: ECE 2104

Course Title: Digital Electronics and Logic Circuits Laboratory

Submitted by – Group B-09 (2309050-2309051)

1. Motivation

Background & Motivation

An asynchronous arbiter is a critical digital circuit used to determine the priority of events that occur nearly simultaneously. In modern computing, this logic is fundamental to interrupt controllers and bus arbitration. The "Reaction Time Race Game" implements this logic to identify which of three contestants reacts fastest to a stimulus. This project provides hands-on experience with latching, timing, and decoding without the use of microcontrollers, bridging the gap between basic logic gates and complex system design.

Conflicting Requirements

The design must balance extreme sensitivity (to catch millisecond differences between players) with high noise immunity (to prevent false triggers from mechanical switch bouncing). Additionally, the system must remain simple enough for breadboard implementation while providing advanced features like individual score tracking and master timing control.

Contribution

This project contributes a standalone, TTL-logic-based competitive system. It features an automated "Lockout" mechanism, a software-free digital scoring system using 74LS90 counters, and a 555-timer-based master control unit to regulate the game duration.

2. Objectives

- To design a 3-channel asynchronous arbiter using NE555 timers.
- To implement a permanent lockout mechanism using diode-logic.
- To integrate 74LS90 and 74LS47 ICs for individual score marking.
- To control the game "window" using a monostable master timer.
- To analyze the propagation delays and timing constraints of TTL circuits.

3. Methodology & Implementation

System Overview

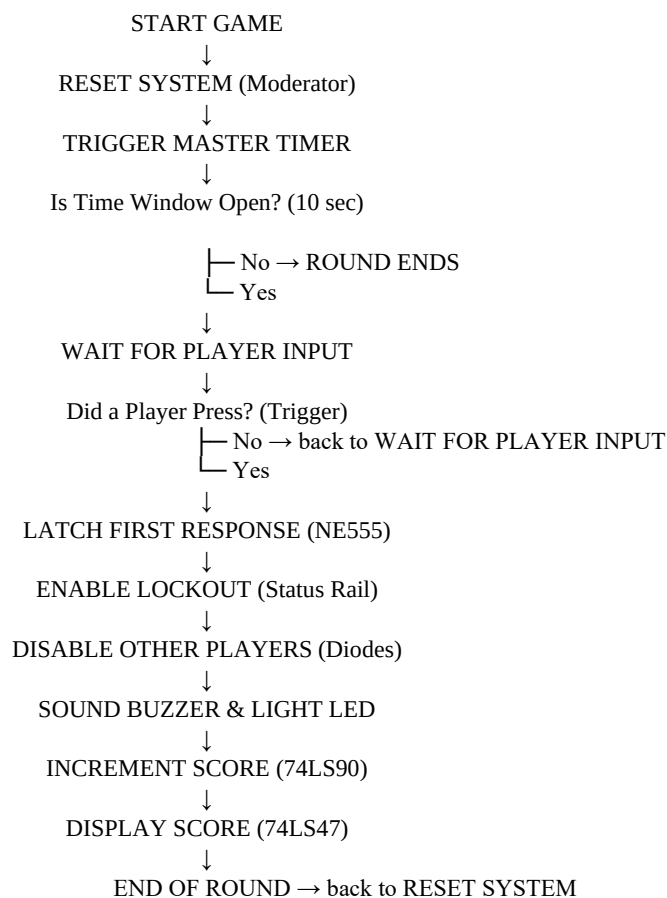
The system operates on a regulated 5V supply and consists of:

- **Input Stage:** Three push-buttons for contestants and one for the moderator.
- **Logic Core:** Three NE555 timers in bistable mode for latching the first-response.
- **Scoring Unit:** Three 74LS90 counters and 74LS47 decoders for score visualization.
- **Control Unit:** A master 555 timer to enable/disable the player rail.
- **Output:** Three 7-segment displays, LEDs, and a piezo buzzer.

Operation Section (System Execution)

1. **Reset:** The moderator clears the scores and latches.
2. **Window Opening:** The moderator triggers the master timer, activating the player buttons for 5-10 seconds.
3. **Capture:** The first player to press their button triggers their NE555. The output goes HIGH.
4. **Lockout:** This HIGH signal passes through a 1N4148 diode to the Status Rail, locking out all other players.
5. **Marking:** The HIGH signal increments the 74LS90 counter. The 74LS47 decoder displays the updated score on the 7-segment display.
6. **Alert:** A buzzer sounds briefly to signal the win.

Flow Chart –



4. Final Outcome Analysis

The expected outcome is a fully functional, zero-latency competition system. The lockout mechanism will ensure that even if two players press within microseconds of each other, only the first is recorded. Using 5V for ICs and 2.2V for displays (via resistors) ensures stable operation as per the lab manual.

5. Cost Analysis

Component	Qty	Unit Cost (BDT)	Total Cost (BDT)
NE555 Timer	5	20	100
74LS90 / 74LS47	6	40	240
7-Segment Display	3	25	75
Passive Components	25	2	50
Total			465 BDT

6 . Real-Life Impact

- **Educational:** Simplifies the understanding of computer architecture interrupts.
- **Professional:** Foundational logic for game show systems and industrial safety interlocks.
- **Environmental:** Uses low-power TTL components and minimal e-waste.

7. Conclusions

The "Reaction Time Race Game" successfully demonstrates the practical application of asynchronous sequential logic and timing synchronization. By utilizing NE555 timers as bistable latches, the system achieves a high-precision arbitration mechanism that effectively manages conflicting input signals without the need for a microcontroller. The integration of 74LS90 decade counters and 74LS47 decoders proves that automated data tracking and visual feedback can be implemented reliably using discrete TTL components. This project serves as a foundational exercise in understanding propagation delays, diode-logic lockout systems, and regulated power management within digital architectures.